sampling will become inconsistent: the force will be dominated by the stochastic term, leaving the dynamics to evolve as a random walk.

Similar behavior is exhibited using the splitting strategy in (7.5), when interposing the *kick* term B between two O steps, such as in the schemes ⟦AOBOA⟧ or ⟦BOAOB⟧. By avoiding "OBO" updates, we can derive schemes that are consistent even at infinite friction (the ⟦BAOAB⟧ and ⟦ABOBA⟧ schemes are examples of such methods).

The Brünger-Brooks-Karplus (BBK) Langevin integrator [55] is one of the most popular and widely-implemented schemes.

Brünger-Brooks-Karplus (BBK)

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= (1 - h\gamma/2)\boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}_n \\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2} \\
\boldsymbol{p}_{n+1} &= [\boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}'_n]/(1 + h\gamma/2)
\end{aligned}$$

Often $\mathbf{R}'$ is chosen such that $\mathbf{R}'_n = \mathbf{R}_{n+1}$, so the random vector would be reused at the start of the following timestep, however there are many variants of this scheme that incorporate the redraw of random numbers differently. In other variants of BBK we take $\mathbf{R}'_n = \mathbf{R}_n$, reusing the random number within the step. We may also consider independent $\mathbf{R}'_n$ and $\mathbf{R}_n$, however when using independent noise processes we must modify the algorithm as it will not sample canonically (see Exercise 1).

7.4 Error Results for Harmonic Problems

For general systems it can be very challenging to give a complete description of the weak error in averages computed from trajectories evolved through a Langevin dynamics discretization scheme. However, using a linear model problem significantly reduces the complexity, while using small dimensional test problems allows us to effectively annihilate sampling error in demonstrative numerical experiments.

As such, we return now to the harmonic oscillator, which as well as being the simplest molecular model is also one of the most relevant for molecular dynamics applications, as many issues of stability and timestep in molecular dynamics simulations arise due to harmonic potentials used to model covalent bonds (such as in crystalline solids and biomolecules). The canonical distributions of position and momentum are also of a simple form (Gaussians), making such oscillators particularly amenable to analysis.

Recall that the one-dimensional harmonic oscillator with spring constant Ω^2 has potential energy function $U(q) = \Omega^2 q^2/2$, with Hamiltonian

$$H(q,p) = \frac{p^2}{2m} + \frac{\Omega^2 q^2}{2}, \tag{7.6}$$

where $(q,p) \in \mathbb{R}^2$ represent a particle's position and momentum, and $m > 0$ is the particle's mass. In this setting, the canonical distribution ρ_β is the product of two

Gaussian distributions, where we have invariant (stationary) averages

$$\langle q \rangle = \langle p \rangle = 0,$$

and

$$\langle \Omega^2 q^2 \rangle = \langle p^2/m \rangle = k_{\mathrm{B}}\mathrm{T}, \qquad \langle qp \rangle = 0.$$

We expect that the discretization schemes that we employ will introduce errors in the computed values of the invariant (long-time) averages. In the large-time limit, these averages will converge towards their invariant value, which we can find using the procedure outlined in [59].

For the harmonic oscillator, the force $-U'(q)$ will be linear, and therefore we would expect that any update step in an algorithm will be linear in the state variables q and p. Thus, denoting the iteration index as a subscript, we may explicitly write one iteration of a general numerical method for evolving the dynamics as

$$\begin{bmatrix} q_{n+1} \\ p_{n+1} \end{bmatrix} = \Psi \begin{bmatrix} q_n \\ p_n \end{bmatrix} + \mu_n, \tag{7.7}$$

where Ψ is a constant matrix that depends on the stepsize h, the friction coefficient γ, the particle mass m and the spring constant Ω. Additionally μ_n is a vector of stochastic processes. For example, in the BBK method we can compute directly from its update scheme that

$$\Psi^{\mathrm{BBK}} = \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2m} & \frac{h}{m}\left(1 - \frac{\gamma h}{2}\right) \\ \frac{h\Omega^2}{2m}\left(\frac{h^2\Omega^2 - 4m}{2+\gamma h}\right) & \frac{(2-\gamma h)\left(2m - h^2\Omega^2\right)}{2m(2+\gamma h)} \end{bmatrix},$$

$$\mu_n^{\mathrm{BBK}} = \begin{bmatrix} \sqrt{\frac{k_{\mathrm{B}}\mathrm{T}\gamma h^3}{2m}}\mathrm{R}_n \\ \frac{\sqrt{2k_{\mathrm{B}}\mathrm{T}\gamma hm}}{2+\gamma h}\left(\left(1 - \frac{h^2\Omega^2}{2m}\right)\mathrm{R}_n + \mathrm{R}'_n\right) \end{bmatrix}.$$

More generally, if we define the components of Ψ as ψ_{ij} and denote $\mu_n = [\mu_{n,1}, \mu_{n,2}]^T$ we can take products of the update equations defined by (7.7) to obtain

$$q_{n+1}^2 = \psi_{11}^2 q_n^2 + \psi_{12}^2 p_n^2 + \mu_{n,1}^2 + 2\psi_{11}\psi_{12} q_n p_n + 2\psi_{11}\mu_{n,1} q_n + 2\psi_{12}\mu_{n,1} p_n, \tag{7.8}$$

$$p_{n+1}^2 = \psi_{21}^2 q_n^2 + \psi_{22}^2 p_n^2 + \mu_{n,2}^2 + 2\psi_{21}\psi_{22} q_n p_n + 2\psi_{21}\mu_{n,2} q_n + 2\psi_{22}\mu_{n,2} p_n, \tag{7.9}$$

$$\begin{aligned} q_{n+1}p_{n+1} = {} & \psi_{11}\psi_{21} q_n^2 + \psi_{12}\psi_{22} p_n^2 + \mu_{n,1}\mu_{n,2} + (\psi_{11}\psi_{22} + \psi_{12}\psi_{21}) q_n p_n \\ & + (\psi_{11}\mu_{n,2} + \psi_{21}\mu_{n,1}) q_n + (\psi_{22}\mu_{n,1} + \psi_{12}\mu_{n,2}) p_n. \end{aligned} \tag{7.10}$$